

HR EMPLOYEE ATTRITION ANALYTICS

Project Focus	Employee attrition analysis and workforce insights
Primary Tool	Power BI / Python / SQL
Dataset Size	1,470 employee records across 35 fields

1. Project Overview

This project analyzes employee attrition patterns to help a business understand workforce turnover, employee characteristics, and factors that may be associated with attrition. The dashboard is designed for managers, HR teams, and decision-makers who need a concise view of the workforce and areas that may need further investigation.

Business Goal	Identify attrition patterns and workforce segments that may deserve attention.
Analytical Goal	Summarize employee demographics, job characteristics, overtime, satisfaction, and attrition indicators in an interactive dashboard.
Output	Interactive Power BI dashboard with KPI cards, charts, slicers, and a concise reporting view.

Key dashboard KPIs used in the project

Total Employees	1,470
Attrition	237 employees
Attrition Rate	Approximately 16.1%
Average Age	Approximately 36.9 years

2. Data Source and Preparation

The project uses the employee attrition dataset prepared for the analysis. The working dataset contains 1,470 employee records and 35 columns covering demographic, job, compensation, satisfaction, and attrition-related attributes.

Preparation workflow

- Inspected the dataset structure, column names, and data types.
- Checked for missing values, duplicates, and inconsistent values.
- Validated numerical, categorical, and binary fields used in analysis.
- Created calculated measures for dashboard KPIs and comparative views.
- Built report pages and slicers around business questions rather than raw field lists.

Important analytical note

Patterns shown in the dashboard are descriptive. An association between a factor and attrition does not, by itself, establish that the factor caused employees to leave.

3. Analysis and Dashboard Design

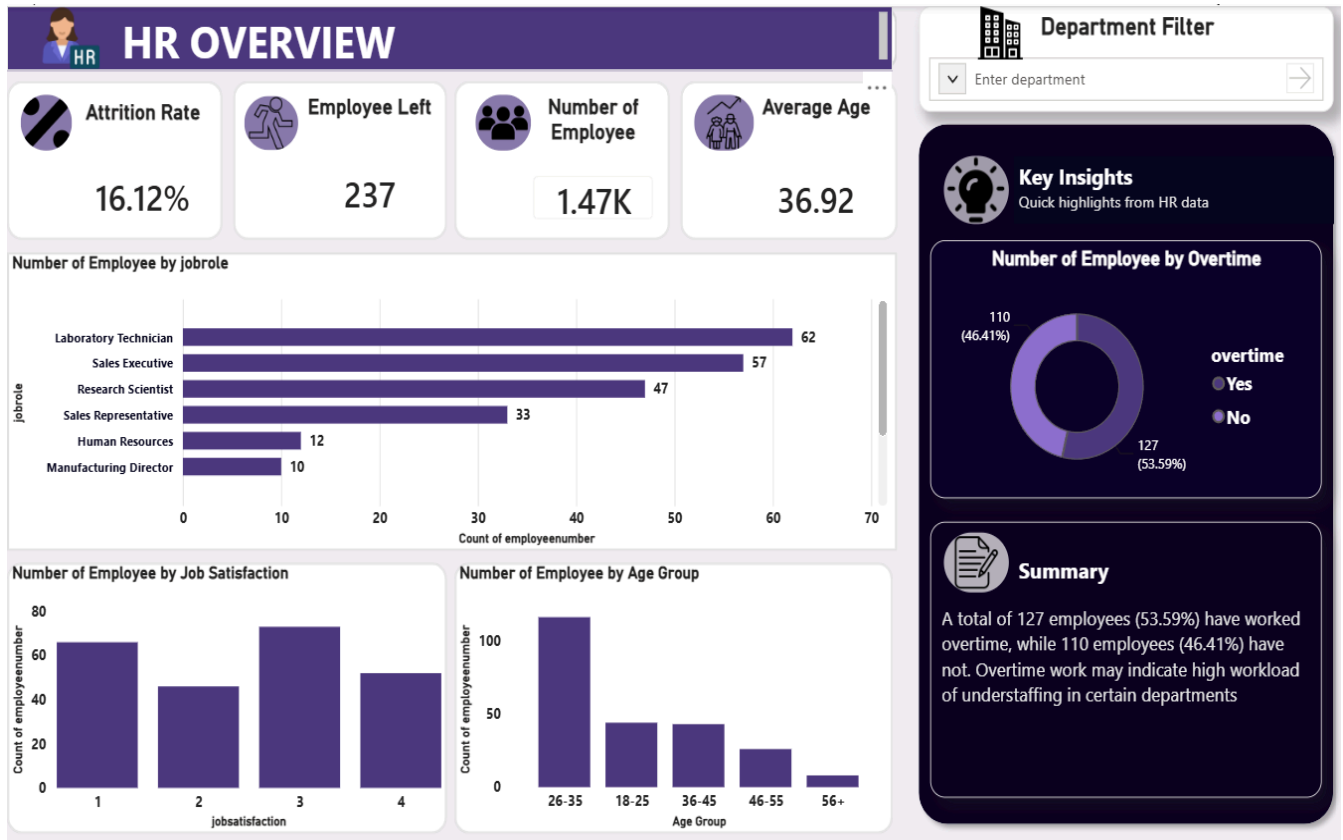
The analysis focuses on understanding where attrition occurs across employee groups and how workforce characteristics differ between employees who stayed and those who left.

Attrition by Role	Compare attrition across job roles to identify roles with visibly different turnover levels.
Attrition by Age	Examine how attrition is distributed across age groups.
Overtime	Compare attrition patterns for employees with and without overtime.
Job Satisfaction	Review satisfaction levels alongside attrition outcomes.
Department Slicer	Allow users to focus the analysis on a specific department.

Example business questions answered

- Which employee groups show the highest number or rate of attrition?
- Which job roles appear to have more turnover and require deeper investigation?
- How does overtime status differ between employees who stayed and those who left?
- How do satisfaction patterns look across the workforce?

4. Dashboard Screenshot



5. Key Insights and Interpretation

The following points can be used as the written interpretation in the portfolio case study. They are intentionally phrased as descriptive findings rather than causal conclusions.

Workforce Scale	The dashboard represents 1,470 employees, giving management a clear view of the overall population.
Attrition Level	237 employees are classified as attrition cases, corresponding to an attrition rate of approximately 16.1% in this dataset.
Age Profile	The average employee age is approximately 36.9 years; age distribution can be explored to identify groups with visibly different attrition levels.
Overtime	The dashboard compares overtime status with attrition, providing a basis for targeted follow-up analysis rather than assuming overtime is a cause.
Job Role	Role-level comparisons can highlight segments where turnover is concentrated and where HR may want deeper qualitative investigation.

6. Recommendations

- Use the dashboard as a monitoring tool and investigate segments with consistently elevated attrition. - Combine dashboard findings with employee feedback, exit interviews, and HR context before taking action.
- Track attrition over time in future versions of the project to distinguish temporary changes from persistent patterns.
- Consider adding tenure, compensation bands, location, and recruitment data for richer workforce analysis.

7. Technical Summary

Analytics Platform	Microsoft Power BI
Data Preparation	Power Query / structured cleaning and validation
Measures	DAX measures for KPIs, percentages, and dashboard calculations
Visualization	KPI cards, charts, slicers, and report-style narrative
Portfolio Use	Project card thumbnail + linked PDF case study

Conclusion

This project demonstrates an end-to-end approach to HR analytics: inspect the data, validate the fields, transform the dataset for reporting, build meaningful measures, visualize the patterns, and translate the results into business-oriented observations. The dashboard is designed to support investigation and communication rather than replace HR judgment.